Model Immunization from a Condition Number Perspective

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Title Illustration Placeholder

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Abstract & Motivation

- Goal: pre-train models that resist fine-tuning on harmful tasks while preserving utility on benign tasks.
- Key idea: analyze and control **Condition Number** of the fine-tuning Hessian.
- Provide a framework for linear models and an algorithm with condition-number regularizers.
- Empirically effective on linear regression/classification and deep nets (ResNet, ViT).

Condition number closer to 1 → easier optimization; larger → slower convergence for gradient-based methods.

Introduction

- Immunization (Zheng & Yeh, 2024): hinder malicious fine-tuning; mitigate misuse of open-source models.
- Prior work: empirical on text-to-image; lacked precise definition and conditions for success.
- Our perspective: **condition number** of Hessian during linear probing determines fine-tuning difficulty.
- Geometric insight: angles between singular vectors of pre-training vs. harmful covariances matter; metric **RIR**.

Contributions

- Framework using **condition numbers** to define and evaluate immunized models.
- Two regularizers: decrease (**R_well**) and increase (**R_ill**) condition numbers with monotonic guarantees.
- Algorithm integrating task loss + dual regularizers; validated on linear and deep models.

Metric	Tasks
RIR ≫ 1	**Linear + Deep**
Relative Immunization Ratio	Regression · Classification · ResNet · ViT

Preliminaries: Condition Number & GD

- **Condition number** $\kappa(S) = \sigma_{max}(S) / \sigma_{min}(S)$ (pseudoinverse when needed).
- For strongly convex L with Hessian **H**: GD convergence slows as **$\kappa(H)$** increases.
- Regularization idea: shape **$\kappa(H)$** of fine-tuning problems to control optimization difficulty.

Formula placeholders for $\kappa(S)$ and GD rate

Transfer Learning via Linear Probing

- Freeze feature extractor **f_θ**; learn linear classifier **h_w** over target data **D**.
- Optimization: minimize task loss (e.g., **$\ell 2$**, **cross-entropy**) over **w**.
- Focus: harmful task **D_H** vs. pre-training task **D_P**.

Linear probing objective placeholder

Definition: Immunized Model (Linear Setting)

- Harder harmful probing: $\boldsymbol{\kappa}(H_H(\theta_I)) \geq \boldsymbol{\kappa}(H_H(I))$.
- Not harder on pre-training: $\boldsymbol{\kappa}(H_P(\theta_I)) \leq \boldsymbol{\kappa}(H_P(I))$.
- Maintain **D_P** performance; for invertible **θ_I**, equality achievable.

$H(\theta)$ $= \theta^T K \theta$; **K** $= X^T X$ covariance of data.

Analysis: Hessian Spectrum & Alignment

- For linear probing with **$\ell 2$** loss: Hessian **H_H** $= \theta^T K_H \theta$.
- Singular values depend on alignment of **θ** and **K_H** singular vectors (Proposition 3.2).
- Trade-off: if **K_P** and **K_H** singular vectors align, simultaneous improvement is limited.

Diagram placeholder: angles between singular vectors (K_P vs K_H)

Objective and Algorithm (Alg. 1)

- Minimize: task loss **L** + λ_P **R_well**(H_P(θ)) + λ_H **R_ill**(H_H(θ)).
- Update **ω** by ∇_ω L; update **θ** by gradients of **L**, **R_well**, **R_ill**.
- Design ensures monotonic **κ** control under conditions.

Pseudocode Sketch

```
for t = 0..T-1:
  ω ← ω − η ∇_ω L(D_P; ω, θ)
  θ ← θ − η (∇_θ L + λ_P ∇_θ R_well(H_P)
                   + λ_H ∇_θ R_ill(H_H))
```

Implementation Tip

Single backward pass via a dummy layer that left-multiplies grads by inverse covariances.

Flow Diagram of Training Loop

D_P (X_P,Y_P)	→	f_θ → g_ω → L(D_P)
D_H (X_H)	→	f_θ → features → H_H(θ)
Covariance K_P	→	$H\_P(\theta)=\theta^T K\_P\theta$
R_well(H_P)	⇣	∇_θ term (makes κ small)
R_ill(H_H)	⇣	∇_θ term (makes κ large)

ω-update: ∇_ω L

θ-update: ∇_θ L + λ_P∇R_well + λ_H∇R_ill

Condition-Number Regularizers

R_well(S): minimize κ(S)

- Differentiable; upper bounds **log κ(S)**; monotonic κ decrease under GD.
- Apply to **H_P(θ)** to keep benign task well-conditioned.

R_ill(S): maximize κ(S)

- Nonnegative; upper bounds **1/log κ(S)**; differentiable if **σ_min** unique.
- Monotonic **κ** increase with proper step size.
- Apply to **H_H(θ)** to hinder harmful fine-tuning.

For $\mathbf{H(\theta)=\theta^T K\theta}$, gradients w.r.t. **θ** are available; guarantees preserved (Theorems 4.2–4.3).

Theoretical Guarantees

- Monotonic **κ** decrease for **H_P** under **R_well** updates.
- Monotonic **κ** increase for **H_H** under **R_ill** updates.
- Conditions: uniqueness of extreme singular values; appropriate step sizes.

Guarantee illustration placeholder (κ over iterations)

Results: Linear + Deep Nets

Main takeaway: Ours achieves RIR ≫ 1, making harmful probing harder while keeping pre-training well-conditioned.

- **Headline numbers**:
  - **House Prices**: **RIR ≈ 356.2** (Opt κ ≈ 92.6; R_ill Only ≈ 1.24; IMMA ≈ 2.00).
  - **MNIST** (avg over 90 pairs): **RIR ≈ 70.0** with low variance (Opt κ ≈ 69.7 high var).
  - **Deep nets** (ImageNet→Cars/Country211): strong **RIR_θ0**; ImageNet accuracy maintained (ResNet slight drop; ViT improves).
- Convergence behavior: Ours slows **D_H** and speeds **D_P** under exact line search (norm-ratio), consistent with κ control.

Deep Nets: Setup and Metric

- **D_P**: ImageNet; **D_H**: Stanford Cars, Country211.
- Architectures: **ResNet18** (last 2 blocks), **ViT** (final block).
- **$RIR_{\theta 0}$** compares κ vs initialization; $\hat{H}$ from features **f_θ**.

Equation placeholders for $RIR_{\theta 0}$ and $\hat{H}(\theta)$

Fine-tuning Behavior on D_H (Cars)

- Linear probing accuracy vs epochs: **Ours** shows slowest convergence on **D_H**.
- Consistent with higher **κ** on harmful task.

Figure 3 placeholder: Test accuracy vs fine-tuning epochs

Discussion and Limitations

- Guarantees rely on unique extrema and step sizes; combined gradients may interact.
- Linear-theory gap for deep nets; empirical results are positive.
- Hyperparameter balance between **κ(H_P)** and **κ(H_H)** is important.
- Efficient implementation via graph trick; single backward pass.

Notes
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Related Work

- AI safety and model misuse mitigation; unlearning vs immunization.
- Condition number in optimization; preconditioning; recent **κ**-regularization.
- Our novelty: principled **κ**-increase regularizer and immunization framing.

Context
Related Slide 16

Conclusion

- **Condition-number** view yields precise immunization definition and metrics.
- Dual regularizers steer **κ** for harmful vs benign tasks with guarantees.
- Effective across linear models and deep nets while preserving utility.

Code
github.com/amberyzheng/model-immunization-cond-num

Takeaway
RIR ≫ 1

References (Selected)

Boyd & Vandenberghe (2004); Bubeck (2015); Golub & Van Loan (1996)

Nenov et al. (2024) – κ-minimizing regularizer

Zheng & Yeh (2024, 2025) – Model immunization (IMMA baseline; multi-concept extension)

He et al. (2016) ResNet; Dosovitskiy (2021) ViT

Deng et al. (2009) ImageNet; Krause et al. (2013) Cars; Radford et al. (2021) Country211

LeCun (1998) MNIST; Montoya & DataCanary (2016) House Prices

References (Baselines & Methods)

- Zheng, A. Y., & Yeh, R. A. (2024). Model Immunization. Empirical formulation used as baseline **IMMA**.
- Nenov, P., et al. (2024). Differentiable regularizer for minimizing matrix condition number (**R_well** inspiration).
- Opt κ: direct optimization of **$\kappa(H_P) - \kappa(H_H)$** as described in paper's baselines.

See paper Sections 4–5 and Tables 1–3 for exact configurations and numbers.